

AI-Powered Personal Grooming Assistant

Prepared by:

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Project: AI Grooming Model (Fine-tuned Chatbot)

GitHub link-

https://github.com/arkaprava181204/Grooming_AI_Chatbot.git

1. Problem Statement

Young people struggle with grooming due to scattered advice; an AI assistant can provide simple, personalized, affordable skincare, haircare, hygiene, and styling guidance.

2. Proposed AI Assistant

The **Groom Tutor** is a lightweight AI assistant fine-tuned with LoRA to provide personalized, affordable advice on skincare, hair, beard care, hygiene, and styling.

3. Target Users

- Students and young professionals looking for quick, judgement-free grooming advice.
- Individuals with limited access to dermatologists, stylists, or grooming consultants.
- Budget-conscious users seeking low-cost or no-cost grooming and presentation tips.
- Anyone who prefers a private, conversational way to ask personal-care questions rather than searching through articles or videos.

4. Functional Requirements

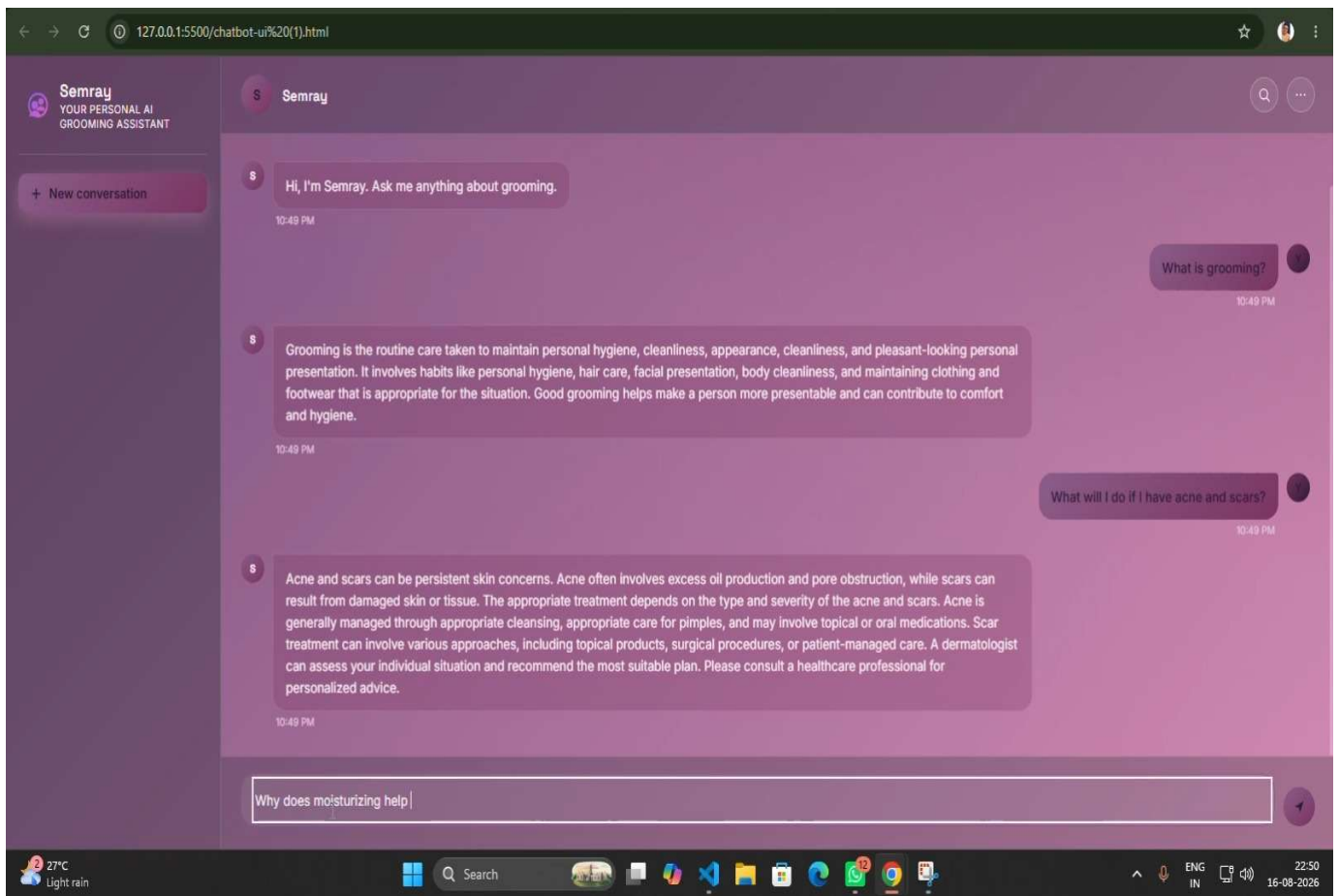
- The system shall accept a natural-language grooming-related question from the user.
 - The system shall generate a relevant, structured, and easy-to-follow response using the fine-tuned language model.
 - The system shall expose the model through an API endpoint so it can be called from a web or mobile front end.
 - The system shall support real-time (or near real-time) response generation.
 - The system shall handle multiple conversational topics: skin care, hair/beard care, hygiene, and affordable grooming.
 - The system shall be deployable and testable without requiring dedicated on-premise GPU hardware (e.g., via Google Colab and a temporary public tunnel).
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5. Key Features

- **Domain-specialised responses:** fine-tuned specifically on grooming and self-care Q&A rather than a general-purpose chatbot.
- **Lightweight base model (Gemma-3 1B-IT):** fast enough to fine-tune and run on a single free-tier GPU (e.g., Colab T4).
- **Parameter-efficient fine-tuning (LoRA):** trains only a small set of adapter weights, reducing compute and storage cost.
- **Fast API access:** chat endpoint that any client application can call.
- **Public accessibility during development:** ngrok tunnelling exposes the local API for quick testing from outside Colab.
- **Structured, step-by-step answers:** responses are broken into categories and bullet points for easy reading.

6. Inputs and Outputs

Type	Description	Example
Input	A free-text grooming-related question from the user, sent as JSON to the API.	"How can I take care of my beard?"
Output	A generated text response from the fine-tuned model, returned as JSON.	Structured tips on beard washing, trimming, and conditioning.
Input (training)	Prompt-completion pairs used to fine-tune the model.	{"prompt": "...", "completion": "..."}
Output (model artifact)	A saved LoRA adapter and tokenizer, later merged with the base model.	./gemma-groom-tutor-1



7. High-Level Solution Design

The solution follows a straightforward fine-tune-then-serve pipeline, illustrated at a high level below:

- **Data Preparation:** A curated set of grooming prompt-completion pairs is assembled and converted into a Hugging Face Dataset, then formatted using the model's chat template.
- **Fine-Tuning:** The base Gemma-3 1B-IT model is fine-tuned using LoRA adapters via TRL's SFTTrainer, targeting the attention projection layers (q_proj, k_proj, v_proj, o_proj).
- **Model Merging:** The trained LoRA adapter is loaded on top of the base model using PEFT, producing a merged grooming-specialised model.
- **Serving Layer:** The merged model is wrapped in a FastAPI application exposing a /chat endpoint, with CORS enabled for cross-origin requests.

- **Public Access:** The local FastAPI server (running via uvicorn in a background thread) is exposed to the internet through an ngrok tunnel for testing and demonstration.
- **Client Interaction:** Any client (web page, mobile app, or test script) sends a POST request with the user's message and receives the generated response.

8. Suggested Technology Stack

Layer	Technology	Purpose
Base Model	Gemma-3 1B-IT (Hugging Face)	Pretrained instruction-tuned language model used as the foundation.
Fine-Tuning	PEFT (LoRA), TRL (SFTTrainer), Transformers	Efficient supervised fine-tuning on domain-specific data.
Training Infra	Google Colab (T4 GPU)	Free/low-cost GPU compute for training and inference.
Backend API	FastAPI	Serves the fine-tuned model over a REST endpoint.
Tunnelling	ngrok	Exposes the local API publicly during development/testing.
Data Handling	Hugging Face Datasets	Structures and formats training data for the trainer.
Client (proposed)	Web or mobile front end (e.g., simple - HTML , css , JS)	User-facing interface that calls the /chat endpoint.

9. Expected Benefits and Impact

- **Accessibility:** Gives users instant, private access to grooming guidance without needing to book a consultation.
- **Cost-effectiveness:** Uses a small, efficiently fine-tuned model, keeping compute and hosting costs low compared to large general-purpose LLMs.
- **Personalisation potential:** A focused, domain-specific model can be further tuned to give more targeted advice than a general chatbot.
- **Learning value:** Demonstrates an end-to-end applied AI workflow — dataset creation, fine-tuning, evaluation, and deployment — valuable for academic and portfolio purposes.
- **Scalability:** The same fine-tuning and serving approach can be extended to other personal-advice domains (fitness, nutrition, study skills, etc.).

10. Future Enhancements

- **Voice-Based Interaction:** Add voice input so users can ask grooming questions by speaking instead of typing.
- **Voice Response:** Enable the assistant to respond through voice, making the interaction more natural and convenient.
- **Video-Based Grooming Analysis:** Allow users to upload or capture a short video for more detailed grooming analysis and recommendations.
- **Real-Time Camera Analysis:** Integrate the camera to provide real-time suggestions related to skin, hair, beard, and overall grooming.
- **Personalized Video Guidance:** Generate step-by-step video guidance showing users how to follow recommended grooming routines.